



Python Tuples

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A tuple is an immutable ordered collection of elements.

- Tuples are similar to lists, but unlike lists, they cannot be changed after their creation.
- Can hold elements of different data types.
- These are ordered, heterogeneous and immutable.

Creating a Tuple

A tuple is created by placing all the items inside parentheses (), separated by commas. A tuple can have any number of items.

```
tup = ()
print(tup)

# Using String
tup = ('Geeks', 'For')
print(tup)

# Using List
li = [1, 2, 4, 5, 6]
print(tuple(li))

# Using Built-in Function
tup = tuple('Geeks')
print(tup)
```

Output

```
()
('Geeks', 'For')
(1, 2, 4, 5, 6)
('G', 'e', 'e', 'k', 's')
```

Creating a Tuple with Mixed Datatypes

Tuples can store elements of different data types, such as integers, strings, lists and dictionaries, within a single structure.

```
tup = (5, 'Welcome', 7.5, True, [1, 2, 3], {'key': 'value'})
print(tup)
```

Output

```
(5, 'Welcome', 7.5, True, [1, 2, 3], {'key': 'value'})
```

Tuple Basic Operations

Accessing of Tuples

We can access the elements of a tuple by using indexing and [slicing](#), similar to how we access elements in a list. Indexing starts at 0 for the first element and goes up to n-1, where n is the number of elements in the tuple. Negative indexing starts from -1 for the last element and goes backward.

```
tup = tuple("Geeks")
print(tup[0])
print(tup[1:4])
print(tup[:3])

# Tuple unpacking
tup = ("Geeks", "For", "Geeks")

# This line unpack values of Tuple1
a, b, c = tup
print(a)
print(b)
print(c)
```

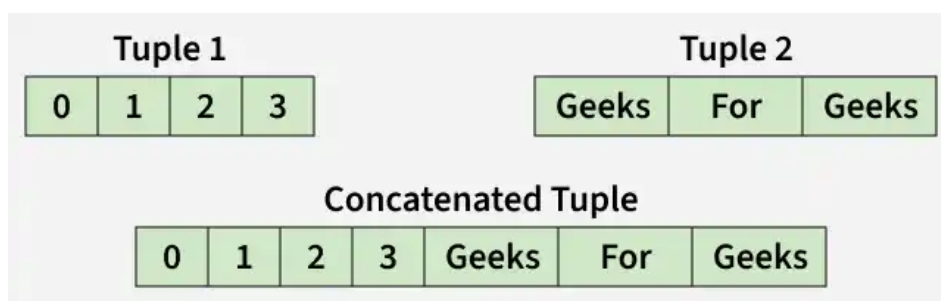
Output

```
G
('e', 'e', 'k')
('G', 'e', 'e')
Geeks
For
Geeks
```

Concatenation of Tuples

Tuples can be concatenated using the + operator. This operation combines two or more tuples to create a new tuple.

Note: Only tuples can be concatenated with tuples. Combining a tuple with other types like lists will raise an error. Tuples themselves can contain mixed datatypes.



concatenation of two tuples

```
tup1 = (0, 1, 2, 3)
tup2 = ('Geeks', 'For', 'Geeks')
tup3 = tup1 + tup2
print(tup3)
```

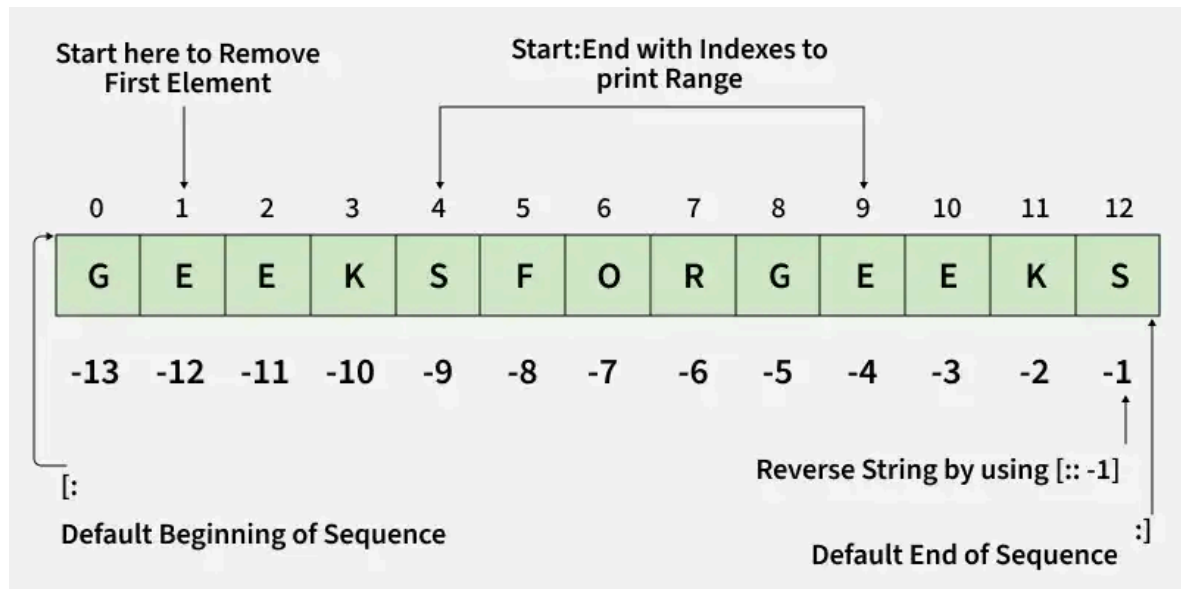
Output

```
(0, 1, 2, 3, 'Geeks', 'For', 'Geeks')
```

Slicing of Tuple

[Slicing a tuple](#) means creating a new tuple from a subset of elements of the original tuple. The slicing syntax is tuple[start:stop:step].

Note: Negative Increment values can also be used to reverse the sequence of Tuples.



tuple slicing

```
tup = tuple('GEEKSFORGEEKS')
print(tup[1:])
print(tup[::-1])
print(tup[4:9])
```

Output

```
('E', 'E', 'K', 'S', 'F', 'O', 'R', 'G', 'E', 'E', 'K', 'S')
('S', 'K', 'E', 'E', 'G', 'R', 'O', 'F', 'S', 'K', 'E', 'E', 'G')
('S', 'F', 'O', 'R', 'G')
```

Note: [:] returns a shallow copy of the tuple, while [::step] allows stepping through elements. Using [::-1] reverses the sequence.

Deleting a Tuple

Since tuples are immutable, we cannot delete individual elements of a tuple. However, we can delete an entire tuple using [del statement](#).

Note: Printing of Tuple after deletion results in an Error.

```
tup = (0, 1, 2, 3, 4)
del tup
```

```
print(tup)
```

ERROR!

Traceback (most recent call last):

File "<main.py>", line 6, in <module>

NameError: name 'tup' is not defined

Tuple Unpacking with Asterisk (*)

*operator is used in tuple unpacking to grab multiple items into a list. This is useful to extract just a few specific elements and collect the rest together.

```
tup = (1, 2, 3, 4, 5)
a, *b, c = tup
print(a)
print(b)
print(c)
```

Output

```
1
[2, 3, 4]
5
```

Explanation: **a** gets the first item, **c** gets the last item and ***b** collects everything in between into a list.

Related Links:

- [Reversing a Tuple](#)
- [Convert a list of Tuples into Dictionary](#)
- [Namedtuple](#)

Suggested Quiz

🔄 10 Questions

How to create an empty tuple in Python?

- ☐ A tup = ()
- ☐ B tup = tuple()
- ☐ C tup = []
- ☐ D Both A and B

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